

Erdős Problem #130

EP-130

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{R}^2$ be an infinite set which contains no three points on a line and no four points on a circle. Consider the graph with vertices the points in A , where two vertices are joined by an edge if and only if they are an integer distance apart.

How large can the chromatic number and clique number of this graph be? In particular, can the chromatic number be infinite?